Whose Writing Is This, Anyway?: Lessons from Building an Open-Source Citation Hallucination Detector

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Abstract

I describe the development and evaluation of an open-source tool for detecting hallucinated citations in academic papers, built entirely on free public APIs (CrossRef, OpenAlex, Semantic Scholar). Testing against 391 real citations and 500 fabricated ones, I find that deterministic validation achieves 0% false-positive rate and 100% detection on citations with DOIs, but is nearly blind to fabricated citations without DOIs (4% detection). Adding AI analysis (Gemini 2.5 Flash, free tier) raises detection to 94% on fakes, but introduces a 72% false-positive rate on real arXiv preprints.

The central finding is not about the tool. It is about an epistemological error that recurs at every tier of detection: the conflation of unverifiable with fabricated. I identified and fixed this error in my deterministic logic, only to discover that AI models reproduce it independently. Grounded AI's analysis for Nature shows the same pattern — a 22% false-flag rate on their most suspicious publications.

I argue that citation hallucination detection is fundamentally a problem of epistemology, not engineering, and that any system, whether rule-based, AI-powered, or human, must grapple with the distinction between absence and fabrication. The tool, code, data, and all experimental results are freely available.

Keywords: citation hallucination, generative AI, open-source tools, scientific integrity, epistemology, BibTeX validation

1 Introduction

On April 2, 2026, *Nature* published a feature article reporting that 2–6% of papers accepted at major computer science conferences in 2025 contained "hallucinated" citations—references that look plausible but point to papers that do not exist [?]. The authors discuss an exclusive analysis, conducted in collaboration with the commercial firm Grounded AI, that estimated that more than 110,000 of the 7 million scholarly publications from 2025 contained invalid references generated by artificial intelligence.

The article was thorough, well-sourced, and alarming. It was also, in several respects, ironic.

Nature partnered with Grounded AI to analyze over 4,000 publications using a proprietary tool called *Veracity*. Neither the tool, the dataset, the risk-scoring model, nor the 20,000 synthetic papers used to calibrate it were made publicly available. The methodology is described at a high level—enough to understand the approach, not enough to reproduce it. For an article about threats to scientific integrity, the absence of reproducible methods is striking.

The proposed solutions are similarly constrained. The article describes several detection tools: Grounded AI's *Veracity* (commercial), GPTZero's screening tool (commercial partnership with ICLR), Frontiers' in-house AI (institutional), and CNRS's *bibCheck* (available only to CNRS scientists). The only open-source option mentioned—*CheckIfExist* [?]—receives a single sentence. The implicit message is that detection requires either money or institutional affiliation. Yet the researchers most vulnerable to hallucinated citations—editors of small journals, independent reviewers, scholars at under-resourced institutions—are precisely the ones who lack both.

This framing obscures a crucial fact: the infrastructure to solve this problem already exists, for free. CrossRef indexes over 150 million DOIs [?]. OpenAlex covers 250 million scholarly works [?]. Semantic Scholar, arXiv, and DataCite are all free and public [?]. The *Nature* article itself relies on checking references against these same databases. The expensive part was never the data—it was the assumption that you need AI to interpret it.

This project started the same week that article was published. I'm the editor of a small mathematics education journal—a venue with no budget for Grounded AI, no institutional screening tools, and no defense against the problem *Nature* described. The question was immediate and practical: *what can I build, right now, with what's freely available?*

The answer, it turned out, was quite a lot. A deterministic validation pipeline using only free public APIs achieves a 0% false-positive rate on 391 real citations and 100% detection on fabricated citations that carry DOIs. No machine learning, no API keys, no subscription fees. The tool is limited—it is nearly blind to fabricated citations *without* DOIs (4% detection)—but what it reveals about the nature of that limitation may matter more than the tool itself.

Adding AI analysis (Gemini 2.5 Flash, free tier) raises detection to 94% on fakes without DOIs. But it also introduces a 72% false-positive rate on real arXiv preprints. The AI makes the same error I had already found and fixed in the deterministic logic: it treats "I cannot verify this citation" as "this citation is probably fake." Grounded AI's own analysis for *Nature* shows the same pattern—22 of their 100 most-flagged publications turned out to be genuine.

This recurring error is the central finding of this paper. It is not primarily a technical problem. It is an epistemological one: the conflation of *unverifiable* with *fabricated*, of absence of evidence with evidence of absence. I encountered it in my own code, watched an AI reproduce it independently, and found it reported at industrial scale in the very article that motivated this work. Any detection system, whether rule-based, AI-powered, or human, must grapple with this distinction. Most do not.

The remainder of this paper describes the tool and its evaluation (Methods), presents experimental results across deterministic and AI-assisted detection tiers (Results), develops the epistemological argument and its practical consequences (Discussion), and identifies the limitations that bound the claims (Limitations and Future Work). The complete codebase, all test data, and all experimental results are publicly available at https://github.com/OhioMathTeacher/Ohio-Journal-of-School-Mathematics.

2 Related Work

The Nature feature by ?] brought citation hallucination to broad attention, but the underlying research had been building for months. This section surveys the empirical studies that document the problem's scope, the detection tools that have been proposed, and the datasets available for evaluation.

2.1 Scope of the Problem

Several independent studies converge on similar estimates. ?] compared high-performance computing conference papers from 2021 and 2025, finding 0% hallucinated citations in the earlier year and 2–6% in the later—a clean before-and-after picture of AI-assisted writing's impact. ?] tracked hallucinated citations across three major NLP venues (ACL, NAACL, EMNLP) and found growth from 20 affected papers in 2024 to 275 in 2025. ?] examined 53 accepted NeurIPS 2025 papers and extracted 100 fabricated citations that had evaded 3–5 expert reviewers per paper, providing a taxonomy of deception strategies: stolen DOIs, plausible fakes, Frankenstein citations (real authors paired with fabricated titles), and others. My evaluation datasets draw directly on this taxonomy.

?] tested LLM citation accuracy across four scientific domains and found that only 50.9% of LLM-generated BibTeX entries were fully correct—meaning roughly half contained at least one fabricated or erroneous field. Their 931-paper benchmark provides field-level error annotations, enabling finer-grained evaluation than citation-level binary classification.

On the ethical and policy side, **?**] argued that hallucinated citations constitute research misconduct when citations function as data—as in systematic reviews, meta-analyses, or literature surveys where fabricated references corrupt the evidence base, not just the bibliography.

2.2 Detection Tools

The detection landscape divides sharply between commercial and open-source approaches.

On the commercial side, Grounded AI's *Veracity* tool—used in the Nature analysis—screened over 4,000 publications using a proprietary risk-scoring model calibrated against 20,000 synthetic papers. Neither the model, the calibration data, nor the tool itself were made public. GPTZero partnered with ICLR for conference-level screening, and Frontiers developed in-house AI detection—both institutional solutions unavailable to independent researchers or small journals.

On the open-source side, **?**] released *CheckIfExist*, which queries CrossRef, Semantic Scholar, and OpenAlex to verify citations—the same databases my tool uses. CheckIfExist is the closest comparable work and the only open-source tool mentioned in the Nature article, where it received a single sentence. **?**] analyzed 2.2 million real citations from 56,381 papers at scale, finding that 1.07% contained invalid references, and released the *CiteVerifier* framework. Separately, **?**] introduced *CiteAudit*, a human-validated benchmark of 9,442 citations (6,475 real, 2,967 fake) drawn from Google Scholar, OpenReview, arXiv, and BioRxiv—the most comprehensive labeled corpus released to date.

My tool differs from CheckIfExist primarily in its status taxonomy—the four-level classification (valid, warning, suspicious, invalid) and the deliberate decision not to treat "not found" as evidence of fabrication. It differs from the commercial tools in being fully open-source, zero-cost, and reproducible: every prompt, dataset, and experimental result is published.

2.3 Datasets and Benchmarks

Evaluation in this space has been hampered by the absence of standard benchmarks. Most studies generate their own synthetic fakes or hand-curate small samples. Three recent datasets offer progress toward shared evaluation:

- **?**]: 100 real hallucinated citations from NeurIPS 2025 papers, with a five-category deception taxonomy. I use this dataset directly.
- **?**]: 931 papers across four scientific domains with field-level error annotations (title, author, year, venue, DOI).
- **?**]: 2.2 million citations from 56,381 papers in AI/ML and Security venues (2020–2025), with 1.07% flagged as invalid.
- **?**]: 9,442 human-validated citations (6,475 real, 2,967 fake) spanning Google Scholar, OpenReview, arXiv, and BioRxiv.

My evaluation uses the Ansari dataset for real-world hallucinations and four additional synthetic categories (Section 3). Running the tool against the Rao [**?**] and CiteAudit [**?**] benchmarks is identified as future work.

3 Methods

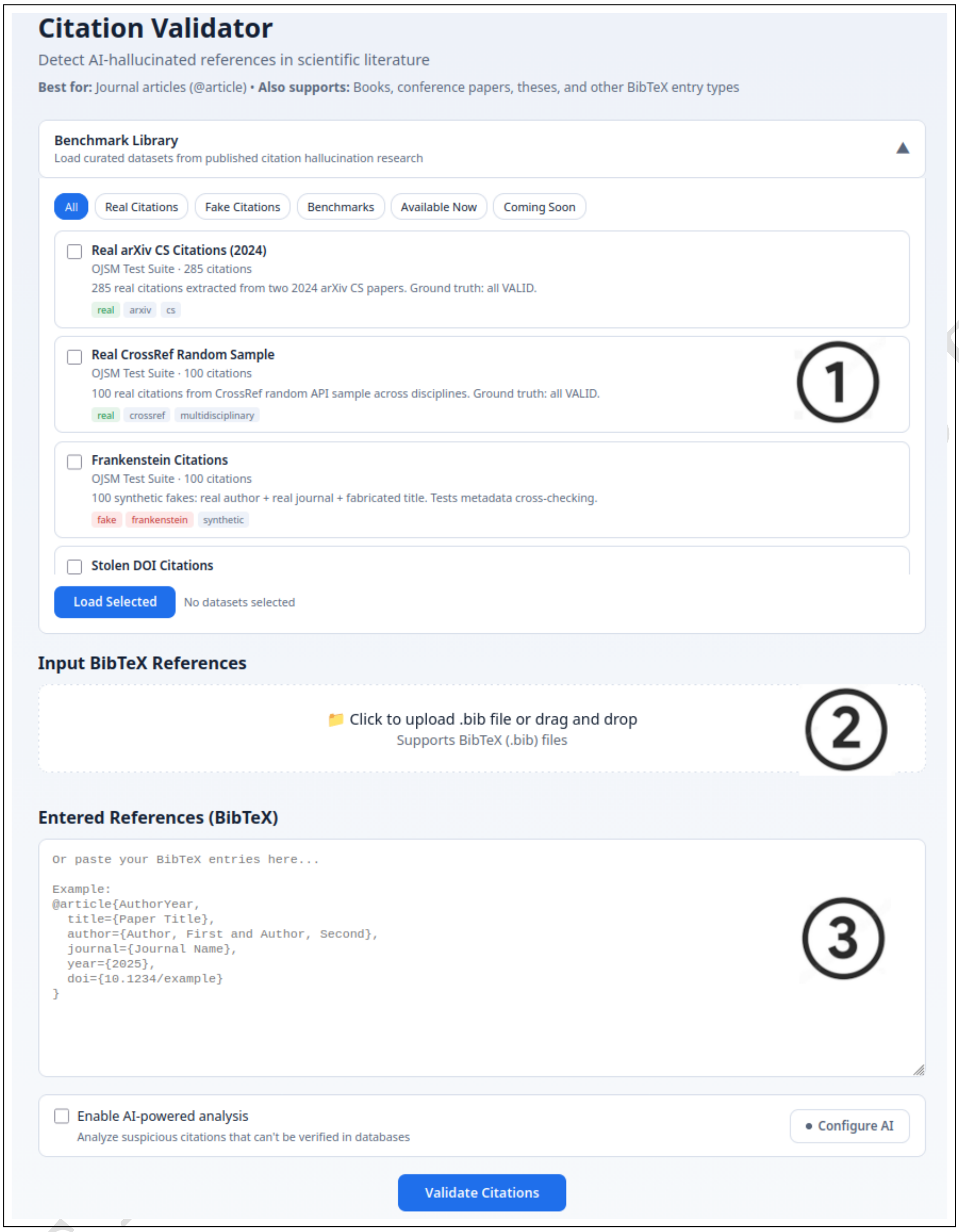
3.1 Design Goals

The tool was designed under three constraints: (1) zero cost to the end user, (2) no proprietary dependencies, and (3) deterministic, auditable results by default, with AI analysis available as an optional second pass. These constraints reflect the intended audience: editors of small journals, independent reviewers, and educators at under-resourced institutions who lack budgets for commercial screening tools.

3.2 Validation Pipeline

The *Citation Validator* accepts a BibTeX bibliography (pasted into a web interface or passed to a command-line script, Figure 1) and classifies each citation into one of four statuses:

Figure 1: *The* Citation Validator *web interface.*



Note: *The interface provides three input methods.* ① ***Benchmark Library:*** *Pre-loaded datasets of real and fabricated citations, organized by category and filterable by type. Users can load curated test sets to evaluate the validator or explore known hallucination patterns.* ② ***File Upload:*** *Drag-and-drop or click to upload a* `.bib` *file directly.* ③ ***Text Entry:*** *Paste raw BibTeX entries into the editor. An optional AI analysis toggle (bottom of interface) enables Tier 3 processing with a configurable provider. The tool is freely hosted at https://huggingface.co/spaces/ojsm/citation-validator.*

Each citation is classified into one of four statuses: *valid* (verified against databases), *warning* (unverifiable but not flagged), *suspicious* (multiple red flags; flagged), or *invalid* (confirmed fake or failed DOI; flagged). Only *suspicious* and *invalid* citations count as detections. The *warning* status is deliberately not treated as a flag—this distinction is central to the epistemological argument developed in the Discussion.

Validation proceeds through three tiers, each progressively more aggressive. Most citations are resolved by Tier 1 alone.

3.2.1 Tier 1: DOI Resolution

A DOI—a Digital Object Identifier—is a permanent link assigned to a published work. When a citation includes one, it is the most reliable piece of metadata available: legitimate DOIs do not expire, and each one points to exactly one paper. Tier 1 exploits this by asking, in effect, *does this DOI actually lead where the citation claims it does?*

The tool first queries CrossRef, the largest DOI registry, which covers over 150 million publications. CrossRef returns the official metadata for that DOI: the real title, authors, and year on record. If CrossRef has no record of the DOI at all—the digital equivalent of a phone number that doesn't exist—that is strong evidence of fabrication. If CrossRef does find it but the metadata conflicts with the BibTeX entry (different title, different authors), that is evidence of a "stolen DOI": a real identifier attached to a fabricated citation.

If CrossRef cannot be reached or returns an error, the tool tries two backup services in sequence—DOI.org and its handle API—before giving up. Citations from arXiv (a preprint server widely used in computer science and mathematics) are handled separately, since their DOIs follow a distinct format and arXiv maintains its own metadata database. All of the services queried are free and public; their addresses are listed in Table 1.

Table 1: *Public APIs used by the* Citation Validator.

Service	Endpoint	Purpose
CrossRef	`api.crossref.org/works/`	DOI resolution
DOI.org	`doi.org/api/handles/`	DOI fallback
arXiv	`export.arxiv.org/api/query`	arXiv metadata
OpenAlex	`api.openalex.org/works`	Title search
Semantic Scholar	`api.semanticscholar.org/graph/v1/`	Title + author search

When a DOI resolves successfully, the returned metadata (title, author, year) is cross-checked against the BibTeX entry. Title similarity is computed using word-level Jaccard similarity [?]:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{1}$$

where A and B are the sets of lowercase alphanumeric tokens extracted from the BibTeX title and the resolved title, respectively. A Jaccard score below 0.4 triggers a title-mismatch warning and potential escalation to *suspicious*. Author matching checks whether any last name from the BibTeX `author` field appears (case-insensitive substring match) in the resolved author list. Year matching requires an exact match. Transient failures (HTTP 429 rate limits, 5xx server errors) trigger one automatic retry after a 0.25-second delay. Only a definitive 404 from CrossRef skips the retry.

3.2.2 Tier 2: Database Search

As previously mentioned, citations without DOIs are searched in two open databases:

- **OpenAlex** [?]: structured title filter query, returning the top result. A Jaccard similarity ≥ 0.5 between the query title and the returned title constitutes a strong match; ≥ 0.3 is noted but not validating.
- **Semantic Scholar** [?]: free-text title search, returning title, authors, year, venue, and external identifiers. A strong match requires both title similarity ≥ 0.5 *and* author/year cross-validation.

If neither database returns a match, the citation is classified as *warning* (unverifiable)—**not** *suspicious*. This is the critical design decision. Many legitimate citations (arXiv preprints, non-English journals, older publications, books) are absent from these databases. Treating absence as evidence of fabrication would produce unacceptable false-positive rates, as the experiments below confirm.

3.2.3 Tier 3: AI Analysis (Optional)

For users who want a second opinion, citations with status *warning* or *suspicious* can be routed to a large language model for analysis. The tool supports four providers:

- Gemini 2.5 Flash (free tier; 1 million tokens/day)
- GPT-4o (paid; OpenAI API)
- Claude Sonnet 4 (paid; Anthropic API)
- Llama 3.3 70B via Groq (free tier; rate-limited)

The AI receives the BibTeX entry fields (type, title, author, year, journal/booktitle, DOI) along with any verified metadata retrieved in Tiers 1–2, and is prompted to assess whether the citation shows signs of fabrication, including internal inconsistency, Frankenstein assembly (fragments from multiple real papers), or metadata that conflicts with verified records. The model returns a structured JSON response: `{is_suspicious, confidence, reason}`. A confidence score ≥ 70 with `is_suspicious: true` escalates the citation to *suspicious*; ≥ 85 escalates to *invalid*.

AI analysis is not enabled by default. The deterministic tiers (1–2) are the recommended pipeline for production use; the rationale for this recommendation is developed in the Results.

3.3 Heuristic Enhancement

In addition to the three-tier pipeline, eight heuristic detectors flag common hallucination patterns as supplementary warnings:

1. Future publication years
2. Generic author names (e.g., "Smith," "et al," "Unknown")
3. Generic title patterns (e.g., "A Study of..." with short titles)
4. Missing critical fields (articles without journal or DOI; proceedings without booktitle)
5. Conflicting venue metadata (both `journal` and `booktitle` present)
6. Sparse metadata (title or author under 5 characters)
7. Malformed URLs
8. Pre-1900 publication years

These heuristics contribute warning signals but do not independently escalate a citation to *suspicious*. They are designed to support the selective AI routing described in the Future Work section.

3.4 Evaluation Framework

3.4.1 Confusion Matrix Convention

I adopt standard detection convention throughout:

Label	Meaning
True Positive (TP)	Fake citation correctly flagged
False Positive (FP)	Real citation incorrectly flagged
True Negative (TN)	Real citation correctly not flagged
False Negative (FN)	Fake citation that slips through

Precision = TP/(TP + FP), Recall = TP/(TP + FN), FPR = FP/(FP + TN), and F1 is the harmonic mean of precision and recall. All metrics are computed *per citation*, not per file.

3.4.2 Datasets

Real citations (391 total). Three sources of known-real citations were used to measure the false-positive rate:

Dataset	Source	Count	Characteristics
arXiv CS (2024)	Extracted from 2 published papers	285	Mostly no DOI
CrossRef random	CrossRef random-works API	96	Mostly with DOI
Nature article	Manual entry	10	Mixed

The arXiv dataset is intentionally adversarial for false-positive testing: most preprints lack DOIs and many are not indexed in OpenAlex or Semantic Scholar. If the validator can avoid flagging these, it can likely handle well-indexed citations.

Fabricated citations (500 total). Five categories of fakes, following the taxonomy of **?**]:

Category	Description	Count	DOI?
Ansari 100	Real NeurIPS 2025 hallucinations	100	Mostly no
Frankenstein	Real author + fabricated title	100	No
Stolen DOI	Real DOI + wrong metadata	100	Yes
Plausible	Entirely fabricated with fake DOIs	100	Yes (fake)
Nonsense	Future years, obvious errors	100	No

The Ansari 100 dataset consists of real hallucinated citations found in 53 accepted NeurIPS 2025 papers [**?**], providing ground truth from citations that evaded 3–5 expert human reviewers per paper. The remaining four categories are synthetic, generated with documented scripts included in the repository.

3.4.3 Reproducibility Pipeline

All experiments reported in this paper were run using two command-line scripts included in the repository. The false-positive baseline (`run_fp_baseline.py`) runs the validator directly against the real-citation datasets, requiring no server:

```
python3 scripts/run_fp_baseline.py --step 1
```

The benchmark runner (`run_benchmark.py`) tests fabricated-citation datasets through the full pipeline, with optional AI analysis via a specified provider:

```
python3 scripts/run_benchmark.py --dataset ansari100 --no-ai
python3 scripts/run_benchmark.py --dataset frankenstein --provider gemini
```

Both scripts load BibTeX files from a dataset manifest (`datasets/manifest.json`), pass them through the validation pipeline, compare each citation's final status against ground-truth labels, and compute detection rates, false-positive rates, and timing. Every run is logged as a structured JSON record to `results/experiments/experiment_log.jsonl`, capturing the dataset, provider, git commit hash, timestamp, per-citation results, and aggregate metrics. This log serves as the primary evidence for the results reported below—any claim in this paper can be traced to a specific, reproducible run.

Figure 2: *Terminal output from a benchmark run against the Ansari 100 dataset.*

```
======================================================================
  BENCHMARK: Compound Deception — NeurIPS 2025 Fakes
  Dataset:   compound-deception-ansari
  AI:        OFF
  Version:   e3be576 (2026-04-10 21:14:14)
======================================================================

Loaded 100 BibTeX entries
Ground truth: 100 per-citation labels loaded

Step 1/1: Deterministic validation...
  Done in 103.7s — 3 valid, 93 warning, 0 suspicious, 4 invalid

======================================================================
  RESULTS
======================================================================
  Valid:       3
  Warning:     93
  Suspicious: 0
  Invalid:     4

  Accuracy:    4.0%
  F1 Score:    7.7%
  TP: 4   TN: 0   FP: 0   FN: 96   Skipped: 0
```

Note: *The script loads 100 fabricated citations from the Ansari dataset, runs each through the deterministic pipeline, and reports per-citation status and aggregate detection metrics. Each run is logged with a git commit hash for full traceability.*

3.4.4 Statistical Methods

Confidence intervals on proportions (e.g., false-positive rate) are computed using the Wilson score interval [?], which provides reliable coverage even for proportions near 0 or 1 with small sample sizes. For k successes in n trials, the $1-\alpha$ Wilson interval is:

$$\frac{1}{1+z^2/n}\left(\hat{p}+\frac{z^2}{2n}\pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}\right) \tag{2}$$

where $\hat{p}=k/n$ and $z=z_{1-\alpha/2}$. All intervals reported use $\alpha=0.05$.

3.5 Code Audit

Before any experiments were run, a thorough code review of the approximately 6,000-line codebase identified six issues, three of them critical:

1. **Inverted confusion matrix.** True positives and true negatives were swapped from standard detection convention. All previously computed accuracy metrics used non-standard definitions.
2. **1,500+ lines of dead code.** An abandoned browser-side validation pipeline remained in production, including a contradictory AI prompt that instructed the model to "assume citations are REAL"—the opposite of the active server-side prompt.
3. **Escalation error.** The original logic treated "not found in any database" as evidence of fabrication, producing 100% false-positive rates on legitimate citations. The validator was committing the same epistemological error it was designed to catch.

After fixes, the codebase shrank by 1,604 lines while gaining a 19-test suite and the evaluation framework described above. The code audit was performed by Claude Opus 4.6 (Anthropic), a large language model—an irony I address in the Discussion.

4 Results

4.1 False-Positive Baseline (Deterministic)

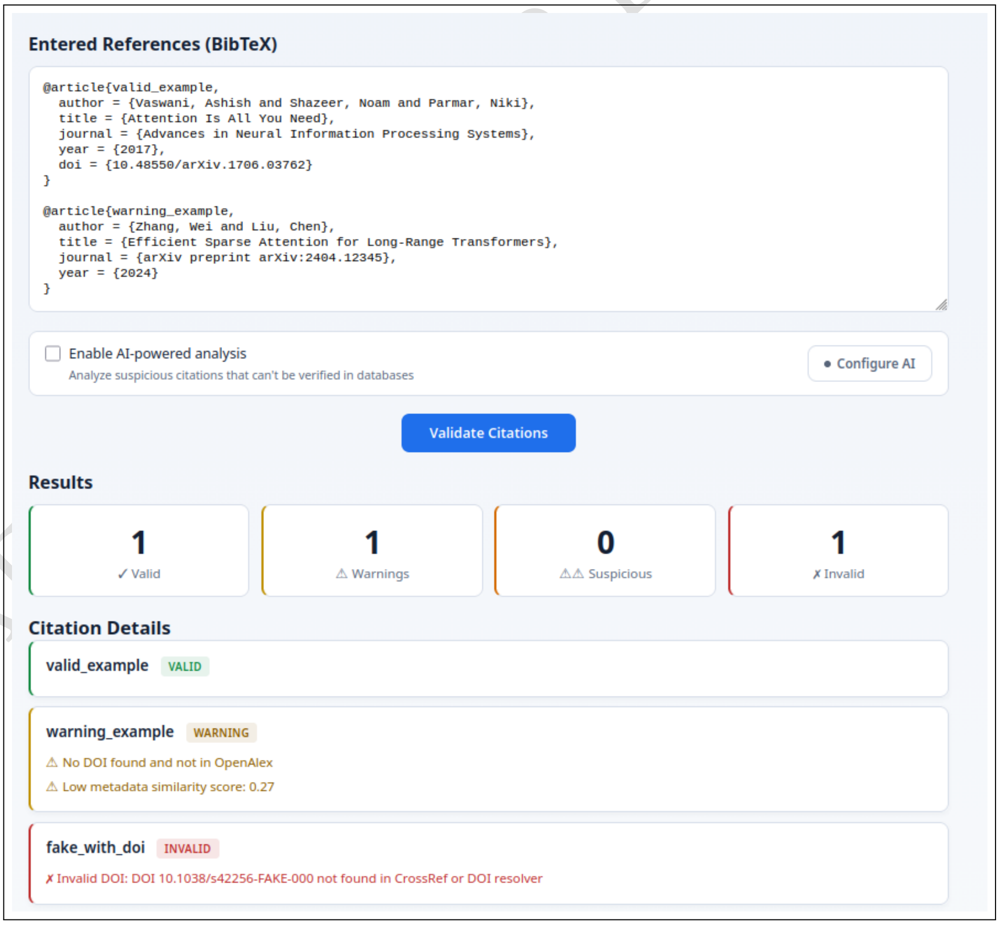
I tested the deterministic pipeline (Tiers 1–2, no AI) against 391 real citations from three sources. Every citation flagged as *suspicious* or *invalid* is a false positive.

Table 2: *False-positive rate on real citations (deterministic validation, no AI).*

Dataset	*n*	Valid	Warning	Suspicious	Invalid	FPR
arXiv CS (2024)	285	2	283	0	0	0.0%
CrossRef random	96	94	2	0	0	0.0%
Nature article	10	5	5	0	0	0.0%
Total	**391**	101	290	0	0	**0.0%**

Zero real citations were flagged as suspicious or invalid. The Wilson 95% confidence interval for 0/391 is [0.0%, 0.96%]—I can claim less than 1% FPR with 95% confidence, but not exactly zero.

Figure 3: Citation Validator *output on a mixed test bibliography.*



Note: *A citation with a valid DOI receives valid status (green). A citation absent from databases receives warning (yellow). A citation with a fabricated DOI receives invalid (red). Only suspicious and invalid statuses count as flags.*

Two features of these results warrant comment. First, 290 of 391 citations (74%) received *warning* status, meaning the validator could not verify them but did not flag them. This is by design: the *warning* category absorbs the uncertainty that would otherwise produce false positives. Second, the arXiv dataset—where 283 of 285 citations landed at *warning*—represents the hardest case for false-positive avoidance. These are legitimate preprints that mostly lack DOIs and are often absent from OpenAlex and Semantic Scholar. The validator's restraint on this dataset is the direct consequence of the escalation fix described in the Methods.

4.2 Fake-Citation Detection (Deterministic)

I tested the same deterministic pipeline against 500 fabricated citations across five deception strategies. Every fake *not* flagged as *suspicious* or *invalid* is a false negative.

Table 3: *Detection rate on fabricated citations (deterministic validation, no AI).*

Category	DOI?	*n*	Detected	Rate
Stolen DOI	Yes	100	100	100.0%
Plausible	Yes (fake)	100	100	100.0%
Ansari 100	Mostly no	100	4	4.0%
Nonsense	No	100	25	25.0%
Frankenstein	No	100	0	0.0%
Total		**500**	**229**	**45.8%**

The pattern is stark: **detection depends entirely on whether the citation has a DOI to check.**

- **With DOI** (Stolen DOI + Plausible): 200/200 = 100%. DOI resolution either fails (fake DOI → *invalid*) or returns mismatched metadata (real DOI + wrong title → *suspicious*).
- **Without DOI** (Ansari + Frankenstein + Nonsense): 29/300 = 9.7%. The validator searches OpenAlex and Semantic Scholar, finds no match, and classifies the citation as *warning*—not *suspicious.*

This is the symmetric consequence of the escalation fix that yielded 0% FPR. The same design decision that prevents flagging legitimate unindexed citations also prevents flagging fabricated ones. The validator correctly says "I cannot verify this" but cannot say "this is fake." This is not a bug. It is the fundamental limit of deterministic detection without DOIs.

The 25% detection rate on Nonsense fakes comes from the heuristic detectors: future publication years and other obvious anomalies that trigger escalation independent of database lookups. The 4% on Ansari 100 reflects the small number of those citations that happened to carry checkable DOIs.

4.3 AI-Assisted Detection

4.3.1 Free Tier: Gemini 2.5 Flash

I ran the Ansari 100 dataset through the full pipeline with Gemini 2.5 Flash as the Tier 3 AI provider. The deterministic tiers ran first; Gemini then analyzed all citations with *warning* or *suspicious* status.

Table 4: *Detection on Ansari 100 fakes: deterministic vs. deterministic + Gemini.*

Pipeline	Detected	Missed	Rate	Cost
Deterministic only	4	96	4.0%	$0
Deterministic + Gemini	94	6	94.0%	$0

Gemini correctly escalated 90 of 96 *warning* citations to *suspicious.* Six fakes remained at *warning* (false negatives). The combined pipeline achieved 94% recall (Wilson 95% CI: [87.5%, 97.2%]) with an F1 score of 0.97, at zero cost via Gemini's free tier (1 million tokens/day). Total token usage was 58,917 tokens (589 average per citation)—well within the daily allowance. At this rate, a journal editor could analyze approximately 1,700 citations per day at no cost.

4.3.2 Paid Tier: Claude Sonnet 4

I ran the identical experiment with Claude Sonnet 4 (Anthropic) as the AI provider, at a cost of approximately \$0.50 per 100 citations. I then ran both providers on a second dataset—the Frankenstein fakes (real authors paired with fabricated titles, no DOIs)—to test whether the Ansari results would replicate.

Table 5: *AI provider comparison across two fabricated-citation datasets.*

Dataset	Provider	Detection	Errors	Time (s)	Cost
Ansari 100	Deterministic	4%	0	94	\$0
	Gemini 2.5 Flash	94%	0	136	\$0
	Claude Sonnet 4	82%	6	429	~\$0.50
Frankenstein 100	Deterministic	0%	0	92	\$0
	Gemini 2.5 Flash	100%	0	120	\$0
	Claude Sonnet 4	100%	0	343	~\$0.50

On the Frankenstein dataset, both providers achieved 100% detection with zero errors. On the Ansari dataset—real hallucinations from published NeurIPS papers, which represent harder and more diverse fakes—Gemini detected 94% (Wilson 95% CI: [87.5%, 97.2%]) versus Claude's 82% ([73.3%, 88.3%]). The intervals overlap: at $n = 100$, the head-to-head difference is not statistically separable at 95% confidence, which is the basis for the cautious framing below. Claude's server errors (6 HTTP 529 responses on Ansari, 0 on Frankenstein) appear to have been transient. Even excluding those errors, Claude's judgment accuracy on successful Ansari calls was 78/90 = 86.7%, below Gemini's 90/96 = 93.75%.

Across both datasets, Gemini was consistently faster (120–136s vs. 343–429s) and had zero errors. The free tier matched or exceeded the paid tier on every metric.

This finding is more robust than a single-dataset comparison, but its limits are real: two datasets ($n = 200$ total), a single prompt per provider, and both datasets consist of fabricated citations without DOIs. The claim is not that Gemini is categorically superior to Claude. The claim is that the free tier is *sufficient*—and that the access barrier imposed by paid APIs is not justified by superior performance on this task.

4.4 The Precision Collapse

The results above show that AI dramatically improves recall on fabricated citations. The critical question is whether it preserves precision on real ones. I ran Gemini on the real-citation datasets from Table 2.

Table 6: *False-positive rate with and without Gemini AI on real citations.*

Dataset	n	Deterministic FPR	+ Gemini FPR
arXiv CS (285 real, mostly no DOI)	285	0.0%	**72.3%**
CrossRef random (96 real, mostly with DOI)	96	0.0%	2.1%

Gemini flagged 206 of 285 real arXiv citations as suspicious—a false-positive rate of 72.3% (Wilson 95% CI: [66.8%, 77.2%]). These are legitimate preprints—the same citations that deterministic validation correctly classified as *warning*. On CrossRef citations, where most entries have DOIs and are well-indexed, Gemini's FPR was only 2.1% (2/96: a Korean-language paper and an 1885 publication).

The contrast is sharp. On fabricated citations, Gemini raises detection from 4% to 94%. On real preprints without DOIs, it raises the false-positive rate from 0% to 72.3%. The weighted FPR across all 381 real citations Gemini analyzed is $(206 + 2)/(285 + 96) = 54.6\%$.

4.5 Summary of Results

Table 7 presents the complete picture.

Table 7: *Summary: what works and what does not.*

What works	What does not (yet)
Deterministic on DOI citations: 0% FPR, 100% detection	Deterministic on non-DOI citations: 0% FPR but only 4–25% detection
Gemini on fakes: 94% detection at $0	Gemini on real preprints: 72% FPR—unusable without selective routing
Free outperforms paid (Gemini > Claude on this task)	AI applied unconditionally destroys precision

The tool is most reliable in deterministic mode: zero false positives, perfect detection on DOI-bearing citations, and honest *warning* on everything else. AI improves recall but requires selective routing—only citations with multiple red flags should be escalated, not every unverifiable citation.

5 Discussion

5.1 The Recurring Error

The central finding of this work is not a detection rate or a cost comparison. It is the observation that a single epistemological error—the conflation of *unverifiable* with *fabricated*—recurs independently at every tier of citation validation I tested.

In the deterministic code. The original escalation logic treated "not found in any database" as evidence of fabrication, producing a 100% false-positive rate on legitimate arXiv preprints. The fix was straightforward: reclassify "not found" as *warning* (unverifiable) rather than *suspicious*. This single change reduced the FPR from 100% to 0%.

In the AI tier. When Gemini 2.5 Flash analyzed the same arXiv preprints, it flagged 72.3% of them as suspicious—reproducing the exact error I had already fixed in the deterministic pipeline. The AI was not given my escalation logic; it arrived at the same mistake independently, from its own training and reasoning. It treated "I do not recognize this paper" as "this paper is probably fake."

In commercial tools. Grounded AI's analysis for *Nature* [?] reported that 22 of their 100 most-flagged publications turned out to be genuine. While their methodology differs from mine, the pattern is the same: a system designed to detect fabrication instead flags legitimate work that happens to be hard to verify.

The error recurs because it is intuitive. If you search for something and cannot find it, the natural conclusion is that it does not exist. But in a world where millions of legitimate publications lack DOIs, are not indexed in major databases, or are published in languages that AI models under-represent, that intuition is systematically wrong. The absence of a database record is information about the *database*, not about the citation.

This has a direct consequence for tool design. Any detection system that treats "not found" as a signal of fabrication—whether through explicit logic, learned patterns, or prompted reasoning—will produce false positives proportional to the incompleteness of its reference data. The only correct response to "not found" is "I don't know," and designing systems that can say "I don't know" without escalating is harder than it sounds.

5.2 The DOI Boundary

The results reveal a sharp discontinuity in detection capability that I call the *DOI boundary*. On one side: citations with DOIs are validated with 100% accuracy and 0% false-positive rate, because DOI resolution provides a ground-truth anchor. On the other side: citations without DOIs are validated with near-zero detection (4% deterministic) or catastrophic false-positive rates (72.3% with AI).

This boundary is not an artifact of the implementation. It reflects a structural feature of the scholarly record: DOIs provide a unique, persistent, resolvable identifier that links a citation to verified metadata. Without that anchor, any validation system—this one or any other—must rely on fuzzy matching against incomplete databases, and fuzzy matching against incomplete databases is where the epistemological error lives.

The practical implication is that the most effective intervention for citation integrity may not be better detection algorithms but broader DOI adoption. Every preprint, dissertation, conference paper, and technical report that receives a DOI becomes trivially verifiable. Every one that does not remains in the zone where detection systems fail or, worse, accuse legitimate scholarship of fabrication.

5.3 Frankenstein Citations

The DOI boundary discussion above assumes the DOI itself is real. Increasingly, that assumption fails. A growing failure mode—one the validator catches immediately—is what I call the *Frankenstein citation*: a reference in which the authors, title, and journal are correct, but the DOI and supporting metadata are fabricated. The paper exists; the identifier does not.

A concrete example encountered during testing: a 2026 journal article cited a 2024 paper with correct authors (Söderström and Palm), correct title, and correct journal (*Research in Mathematics Education*), but with a DOI (`10.1080/14794802.2024.2444321`) that does not resolve in CrossRef and with page numbers (1–23) that differ from the actual pagination (163–184) by more than an order of magnitude. The real DOI was `10.1080/14794802.2024.2401488`. Neither error pattern resembles a human transcription mistake, but together they fit the signature of AI-assisted citation generation: the model correctly identifies that a paper of this description exists, then fabricates the specific identifiers because they were not in its training distribution.

I name this as a category distinct from pure hallucination—which targets papers that do not exist at all—because the detection criteria differ. Frankenstein citations have real bodies but stitched-together identifiers; the validator's metadata cross-check (DOI-title match, year match, author match) catches them, while a tool that only verifies that a DOI exists in CrossRef would not. The Frankenstein category may well prove the more common AI-assisted failure mode in practice. A taxonomy paper applying the validator to a journal's full back catalog is in preparation.

5.4 Free Infrastructure, Public Problem

The *Nature* article frames citation hallucination as a problem requiring sophisticated AI solutions delivered by commercial partners. My results suggest a different framing: the hard part of detection is already solved, for free, by the public infrastructure of scholarly metadata.

CrossRef, OpenAlex, and Semantic Scholar collectively index hundreds of millions of scholarly works and are freely accessible to anyone. A deterministic pipeline querying these databases achieves 0% FPR and 100% detection on DOI-bearing citations—performance that no AI tier in my experiments improved upon. The commercial tools described in the *Nature* article use these same databases; what they add is AI interpretation, which my experiments show introduces more problems than it solves when applied without selective routing.

This is not an argument against AI in citation validation. Gemini's 94% detection rate on fabricated citations without DOIs demonstrates real value. The argument is that AI should be a *second pass* applied selectively to citations that accumulate multiple warning signals—not a wholesale replacement for deterministic verification, and certainly not a service that requires a subscription fee to access.

The cost comparison reinforces this point. Gemini's free tier (1 million tokens/day) outperformed Claude Sonnet 4's paid API (~$0.50/100 citations) on accuracy (94% vs. 82%), reliability (0 vs. 6 errors), and speed (136s vs. 429s). I am cautious about generalizing from a single comparison ($n = 100$), but the direction is clear: for this specific task, the access barrier imposed by paid APIs is not justified by superior performance.

If hallucinated citations are a threat to the shared infrastructure of science—and the evidence suggests they are [**? ? ?**]—then detection tools should be public goods, not proprietary products. The databases are already public. The algorithms are straightforward. The remaining barrier is accessibility: getting a working tool into the hands of people who need it.

5.5 AI as Disease and Cure

This project is layered with contradictions worth naming explicitly. The hallucinated-citation problem is caused by large language models. The code audit that found three critical bugs in this tool was performed by a large language model (Claude Opus 4.6). The AI tier that achieved 94% detection on fabricated citations is itself a large language model (Gemini 2.5 Flash). The same class of technology that poisons citation networks helps make detection tools correct and effective.

I do not pretend this tension is resolved. But I note that refusing to use AI to address AI-generated problems would be like refusing to use a calculator to check arithmetic. The relevant question is not whether AI is involved, but whether its involvement is transparent, auditable, and reproducible. In this project, the AI prompts are published, the model versions are documented, the results are timestamped and stored, and the entire pipeline can be reproduced by anyone with an internet connection and no API keys.

The code audit deserves particular comment. Claude Opus 4.6 identified the inverted confusion matrix, 1,500 lines of dead code, and the epistemological error in the escalation logic—three critical issues I had missed. This is precisely the kind of task where AI adds clear value: systematic review of a large codebase against known correctness criteria. The irony that an AI found the bugs in a tool designed to catch AI-generated errors is not lost on me, but irony is not an argument against effectiveness.

The asymmetry is worth theorizing. Code review and citation validation are not just different tasks; they require opposite epistemic operations. Code review is pattern-matching against a vast corpus of known-correct examples: PEP 8 conventions, standard idioms, type-correct invocations, test patterns. Every well-formed codebase in the model's training data is a positive example of what working code looks like, and bugs are recognizable as deviations from a distribution the model has actually seen. Citation validation reverses the operation. It requires asserting that a paper does *not* exist—that no published work matches this title-author-year combination anywhere in the scholarly record—and there is no positive corpus of "correctly nonexistent" papers to pattern-match against. The model has no calibrated way to distinguish "I cannot recall this paper" from "this paper does not exist," so it falls back on the default heuristic that produced the 72.3% arXiv FPR in this study and the 22 false flags in Grounded AI's analysis: *if I do not recognize it, it is probably wrong.*

This sharpens rather than weakens the argument of §5.1. AI is genuinely useful where it can work *from* known examples, and fails in characteristic ways where the task requires inferring *from absence.* Tools that route every unverifiable citation to AI as a wholesale second pass are asking the model to perform an epistemic operation it cannot calibrate. Tools that route only the multi-flag suspicious cases—using deterministic logic and heuristic detectors to first identify that *something* has gone wrong, and only then invoking AI to characterize *what*—use each mode for what it can actually do. The selective-routing recommendation in the Future Work section is not a tuning preference; it is what the asymmetry between these two cognitive tasks requires.

5.6 The Accessibility Gap

I should be honest about what "free and open-source" currently means in practice.

Using the *Citation Validator* today requires installing Python, running a command-line script or starting a local Flask web server (Figure 1).

For a computer scientist, that is five minutes of setup. For a mathematics education journal editor—which is what I am—it is a real barrier. For a high school teacher evaluating a student's AI-assisted research paper, it is a wall.

The people who most need protection from hallucinated citations are often the least equipped to install developer tools. This is not a new observation—it is the same access gap that makes the commercial solutions described in the *Nature* article inadequate. The commercial tools fail on cost; ours currently fails on technical accessibility. Neither failure is acceptable.

The underlying APIs (CrossRef, OpenAlex, Semantic Scholar) are free and public, but they cannot be called directly from a web browser due to cross-origin security restrictions. A server-side component is required—which means either a local install or a hosted deployment. Hosting is straightforward and free (services like Hugging Face Spaces and PythonAnywhere offer free-tier Python hosting), but someone has to do it.

I name this gap explicitly because I believe it is one of the most important problems this line of work needs to solve. The algorithm works. The APIs are free. The code is open. What is missing is the last mile: getting a working tool into the hands of people who do not know what `pip install` means and should not have to.

6 Limitations and Future Work

6.1 Limitations

Sample size. The false-positive baseline rests on 391 real citations. The Wilson 95% confidence interval for 0/391 is [0.0%, 0.96%]—sufficient to claim less than 1% FPR, but not to claim zero. A single false positive in a larger sample would change the point estimate. The planned 10,000-citation validation study is essential before any stronger claim is warranted.

Domain bias. The real-citation test set is 73% computer science arXiv preprints. Under-represented categories include books, dissertations, non-English publications, retracted papers, pre-1950 works, humanities scholarship, and conference proceedings without DOIs. The validator's performance on these categories is unknown, and there are reasons to expect it may differ: non-English titles may not match well with Jaccard similarity on English tokens, and older works may have incomplete or inconsistent metadata in CrossRef.

Same author, same tests. I wrote the validation code and also designed and curated the test datasets. All code changes addressed general design flaws (escalation logic, confusion matrix convention, dead code removal), not specific test cases. The five test-data corrections in Run 3 (four CrossRef metadata artifacts, one incorrect BibTeX title) were data quality fixes, not validator adjustments. Nevertheless, the risk of unconscious overfitting exists whenever the same person writes both the tool and the evaluation. Independent replication with externally sourced datasets is needed.

AI comparison is preliminary. The Gemini-vs.-Claude comparison uses a single dataset ($n = 100$), a single prompt per provider, and does not control for prompt optimization. Claude's six HTTP 529 errors may reflect transient server load rather than model capability. I report the comparison as a data point, not a definitive ranking of providers.

Synthetic fakes may not represent real hallucinations. Four of the five fabricated-citation datasets are synthetically generated. Only the Ansari 100 dataset contains real hallucinations found in published papers. Synthetic fakes may be systematically easier or harder to detect than real ones, depending on how closely the generation process mimics actual LLM behavior.

No adversarial testing. I have not tested the validator against citations deliberately crafted to evade it. An adversary who understands the pipeline—for instance, one who assigns real DOIs to fabricated metadata or who crafts titles to achieve high Jaccard similarity with existing papers—could likely bypass detection. The tool is designed for the common case (LLM-generated hallucinations in submitted manuscripts), not for adversarial deception.

6.2 Future Work

Selective AI routing. The most pressing technical problem is designing rules for when to send citations to AI analysis. The current approach—route all *warning* citations to AI—produces unacceptable false-positive rates on real preprints (72.3%). A more selective policy would route citations to AI only when multiple heuristic warnings accumulate (e.g., generic author name *and* title pattern *and* not found in databases), while leaving single-warning citations at *warning* status. The eight heuristic detectors already produce these signals; what remains is calibrating the threshold.

Diverse dataset validation. The 10,000-citation study should include: non-English publications (testing Jaccard similarity across languages), books and monographs (testing against WorldCat or similar), dissertations (ProQuest, institutional repositories), humanities and social science journals, retracted papers (Retraction Watch database), and pre-DOI-era publications. Several published benchmarks offer external ground truth: **?**] provide a 931-paper benchmark across four scientific domains with field-level error annotations, and the GhostCite corpus [**?**] includes 2.2 million real citations from 56,381 papers.

Cross-tool comparison. *CheckIfExist* [**?**] is the closest comparable open-source tool. Running identical inputs through both tools would provide a direct performance comparison and might reveal complementary strengths. The CiteAudit benchmark [**?**] offers 9,442 human-validated citations (6,475 real, 2,967 fake) across multiple domains—a rigorous external test set.

DOI-list input mode. The current input methods all assume the user has a BibTeX-formatted bibliography. In practice, many of the people most likely to need citation validation—journal editors checking a manuscript's references, reviewers copying from a PDF, teachers spot-checking a student paper—have a list of DOIs and nothing else. A planned addition is a fourth input mode that accepts one DOI per line and wraps each into a synthetic `@misc` entry before passing it to the existing pipeline. No changes to the validation logic are required; the work is entirely an input-shim, but it materially lowers the barrier for non-LaTeX users.

Hosted deployment. Eliminating the installation barrier is a design goal, not a feature request. The deterministic pipeline requires only server-side Python and free API access. Free hosting platforms (Hugging Face Spaces, PythonAnywhere) can serve the web interface at no cost. A Dockerfile is included in the repository. The goal is a URL that any journal editor or teacher can visit in a browser, paste a bibliography, and receive results—no Python, no terminal, no setup.

Prompt engineering for precision. The AI tier's 72.3% FPR on arXiv preprints may be partly a prompting problem. The current prompt does not explicitly instruct the model to distinguish between "unverifiable" and "fabricated." Prompt modifications that frame the task as a calibrated confidence assessment—rather than a binary suspicious/not-suspicious judgment—may reduce false positives while preserving recall. This is an empirical question that systematic prompt ablation can answer.

User-curated benchmark libraries. The benchmark library currently ships with a fixed set of curated datasets. A natural extension would allow users to save their own entered or uploaded citations as named libraries that persist across sessions and appear in the library dropdown. This would let journal editors build institution-specific test sets—for example, a collection of citations from past submissions that were manually verified or flagged—and share them with colleagues. The underlying architecture (a JSON manifest pointing to BibTeX files) already supports this; what is needed is a save/load interface and lightweight server-side storage.

Integration with editorial workflows. The tool's ultimate value depends on adoption. Zotero and Mendeley plugins, browser extensions, and journal submission system integrations would place validation where editors and reviewers already work. The API-based architecture supports these integrations; what is needed is development effort and community contribution.

7 Conclusion

Citation hallucination is a solvable problem, and solving it should not require an institutional budget or a computer science degree.

The databases that index legitimate scholarship—CrossRef, OpenAlex, Semantic Scholar—are free and public. A deterministic pipeline querying them achieves 0% false-positive rate on 391 real citations and 100% detection on fabricated citations with DOIs, at zero cost. AI analysis can extend detection to citations without DOIs (94% on the Ansari 100 benchmark, also at zero cost via Gemini's free tier), but only with selective routing—applied unconditionally, it produces false-positive rates that would render any tool unusable.

The harder problem is not technical but epistemological. Every detection system I examined—my own deterministic logic, AI models, and the commercial analysis reported by *Nature*—independently made the same error: treating the absence of a database record as evidence of fabrication. I found this error in my code and fixed it. I watched Gemini reproduce it from scratch. I found it reported at industrial scale in Grounded AI's results. It recurs because it is intuitive, and it is dangerous because millions of legitimate publications are absent from the databases these systems query.

Any system that aspires to detect hallucinated citations must first learn to say "I don't know"—and to mean it. The distinction between *unverifiable* and *fabricated* is not a technicality. It is the difference between a tool that helps and a tool that accuses.

As a final check, the bibliography of this paper was itself run through the validator. Of the 13 references, four received *valid* status—the entries with DOIs that resolved cleanly through CrossRef and the DOI handle service—and nine received *warning*—the arXiv preprints and infrastructure references (CrossRef, OpenAlex, Semantic Scholar) that legitimately lack DOIs. Zero references were classified as *suspicious* or *invalid*. The result is, by design, exactly what the argument here recommends: the validator distinguished *verified* from *unverifiable* without conflating either with *fabricated*.

The *Citation Validator*, its complete codebase, all test data, and all experimental results described in this paper are freely available at https://github.com/OhioMathTeacher/Ohio-Journal-of-School-Mathematics. I offer it not as a finished product but as a demonstration: that the infrastructure exists, the problem is tractable, and the tools should be public. I invite the community to test it, break it, improve it, and build on it.

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